**COURSE TITLE : DIGITAL CIRCUITS**
**COURSE CODE : 3212**
**COURSE CATEGORY : B**
**PERIODS/WEEK : 4**
**PERIODS/SEMESTER: 60**
**CREDITS : 4**

## TIME SCHEDULE

| Module | Topics | Periods |
|---|---|---|
| 1 | Number system and Boolean simplifications | 18 |
| 2 | Combinational logic circuits | 15 |
| 3 | Sequential logic circuits | 15 |
| 4 | A/D, D/A, Memories And displays | 12 |
| **TOTAL** | | **60** |

## COURSE OUTCOME

| Module | G.O. | On completion of the study of this module the student will be able |
|---|---|---|
| **1** | 1 | To understand various number system |
| | 2 | To comprehend the conversion |
| | 3 | To perform the arithmetic operations on different number system |
| | 4 | To understand the digital codes |
| | 5 | To comprehend boolean expression simplifications |
| **2** | 1 | To know the logic families |
| | 2 | To comprehend performance characterestics of logic gate |
| | 3 | To understand combinational logic ciruits |
| | 4 | To understand functions of combinational logic |
| **3** | 1 | To understand bistable logic circuits |
| | 2 | To know the types of sequential logic circuits |
| | 3 | To understand counters |
| | 4 | To understand shift register |
| **4** | 1 | To understand analog to digital converter in digital instrument |
| | 2 | To understand digital to analog conveter |
| | 3 | To understand the memory in digital instrument |
| | 4 | To understand the display in digital instrument |

On completion of the units, the student will be able to

**<u>MODULE I</u>** **NUMBER SYSTEM AND BOOLEAN ALGEBRA**

**1.1.0 To understand various number system**
1.1.1 To List the various number system
1.1.2 To illustrate the bases of various number system

**1.2.0 To comprehend the conversion**
1.2.1 To convert decimal to binary, octal and hexadecimal
1.2.2 To convert the binary, octal, and hexadecimal to decimal
1.2.3 To obtain the 2's complement of binary and hexadecimal
1.2.4 To convert binary to hexadecimal and octal
1.2.5 To convert hexadecimal and octal to binary
1.2.6 To express decimal number in the BCD form

**1.3.0 To perform the arithmetic operations on different number system**
1.3.1 To add binary numbers
1.3.2 To add Hexa decimal numbers
1.3.3 To subtract binary numbers
1.3.4 To subtract binary with 2's complement
1.3.5 To subtract Hexadecimal numbers
1.3.6 To subtract Hexadecimal with 2's complement
1.3.7 To multiply binary number
1.3.8 To divide binary number

**1.4.0 To understand the digital codes**
1.4.1 To list various digital codes
1.4.2 To compare weighted and unweighted code
1.4.3 To convert binary to gray and gray to binary
1.4.4 To express the number in Excess 3 format
1.4.5 To express in ASCII

**1.5.0 To comprehend Boolean expression simplifications**
1.5.1 To list Boolean operations
1.5.2 To list Logic Expressions
1.5.3 To realize the operation of Logic gates
1.5.4 To illustrate boolean law
1.5.5 To explain rules for Boolean algebra
1.5.6 To describe Demorgan's theorems
1.5.7 To apply Demorgan's theorems on logic expressions
1.5.8 To list the types of Boolean expressions-pos,sop
1.5.9 To simplify the Boolean expressions using K-map

**<u>MODULE II</u>  COMBINATIONAL LOGIC CIRUITS**

**2.1.0 To know the logic families**
2.1.1 To list the digital logic families
2.1.2 To explain the operation of TTL NAND gate
2.1.3 To explain the operation of TTL NOT gate

**2.2.0 To comprehend   performance characterestics of logic gate**
2.2.1 To define propogation delay
2.2.2 To define power dissipation
2.2.3 To define dc noise margin
2.2.4 To define ac noise margin
2.2.5 To define speed power product
2.2.6 To define fan out
2.2.7 To define fan in

**2.3.0 To understand combinational logic circuits**
2.3.1 To define combinational logic circuits
2.3.2 To implement the logic expression with universal gates

**2.4.0 To understand functions of combinational logic**
2.4.1 To explain the operation of Half adder
2.4.2 To explain the operation of Full adder,
2.4.3 To explain the operation of parallel binary adder
2.4.4 To describe the operation of look-ahead –carry adder
2.4.5 To describe the operation of 4 bit binary decoder
2.4.6 To describe the operation of BCD-DECIMAL decoder
2.4.7 To explain the operation of BCD-7 segment decoder
2.4.8 To describe the operation of Decimal-BCD encoder
2.4.9 To describe the operation of multiplexer
2.4.10 To describe the operation of  Demultiplexer
2.4.11 To describe the operation of comparator
2.4.12 To define glitch, strobing

**<u>MODULE III</u>    SEQUENTIAL LOGIC CIRCUITS**

**3.1.0 To understand bistable logic circuits**
3.1.1 To describe the operation of  S-R, D Latches
3.1.2 To explain the operation of  S-R, D  ,J-K edge triggered flip flops
3.1.3 To illustrate the operation of  S-R, D ,J-K pulse  triggered flip flops
3.1.4 To explain the operation of master slave J-K pulse triggered flip flop
3.1.5 To list asynchronous inputs to flip flop
3.1.6 To list the applications of flip flop

**3.2.0 To know the types of sequential logic circuits**
3.2.1 To define synchronous counter
3.2.2 To define asynchronous counter
3.2.3 To define shift register

3.2.4 To compare synchronous counter and asynchronous counter
3.2.5 To list the applications

**3.3.0 To understand counters**
3.3.1 To design 4 bit asynchronous counter
3.3.2 To design Decade asynchronous counter
3.3.3 To design 4 bit synchronous counter
3.3.4 To design decade synchronous counter
3.3.5 To define the modulus of counter
3.3.6 To explain the up/down asynchronous counter

**3.4.0 To understand shift register**
3.4.1 To Describe serial in serial out
3.4.2 To Describe serial in parallel out
3.4.3 To Describe parallel in serial out
3.4.4 To Explain parallel in parallel out
3.4.5 To Describe applications

**<u>MODULE IV</u> A/D, D/A, MEMORIES AND DISPLAYS**

**4.1.0 To understand analog to digital converter in digital instrument**
4.1.1 To list various analog to digital converters
4.1.2 To explain the operation of single slop A/D converter
4.1.3 To explain the operation of S-A A/D converter
4.1.4 To list some A/D ICs

**4.2.0 To understand digital to analog converter**
4.2.1 To explain the operation of R- 2R D/A converter
4.2.2 To explain the working of binary weighted converter
4.2.3 To list some D/A ICs

**4.3.0 To understand the memory in digital instrument**
4.3.1 To List various types of memory
4.3.2 To define Byte, nibble, capacity
4.3.3 To explain the various RAM
4.3.4 To explain the various ROM
4.3.5 To compare ROM & RAM

**4.4.0 To understand the display in digital instrument**
4.4.1 To List various display in digital meter
4.4.2 To list the specifications of digital meter display
4.4.3 To calculate display digit and counts
4.4.4 To define resolution of digital meter
4.4.5 To define Sensitivity of digital meter

# CONTENT DETAILS

## MODULE I

Various number system-octal-decimal-hexadecimal-bases of various number system- conversion of various number system-arithmetic operations on different number system -weighted and unweighted code - - Logic gates -Boolean law- Demorgan's theorems- Boolean expressions-pos,sop-Boolean expressions simplifications using K-map

## MODULE II

Logic families-the operation of TTL NAND gate-operation of TTL NOT gate- performance characteristics of logic gate-propagation delay-power dissipation-dc noise margin -ac noise margin-speed power product-fan out-fan in Combinational logic circuits- logic expressions with universal gates- half adder-full adder-parallel binary adder-look-ahead –carry adder-4 bit binary decoder-BCD-DECIMAL decoder-BCD-7 segment decoder-Decimal-BCD encoder-multiplexer-Demultiplexer-comparator-glitch-strobing.

## MODULE III

Bistable logic circuits-operation of S-R, D Latches - edge triggered flip flops pulse triggered flip flops -Master slave J-K pulse triggered flip flop- asynchronous inputs to flip flop-applications of flip flop- Types of sequential logic circuits-synchronous counter-asynchronous counter-shift register-applications-compare synchronous counter and asynchronous counter-applications - counters-Design 4 bit asynchronous counter-Decade asynchronous counter-4 bit synchronous counter-decade synchronous counter-the modulus of counter- the up/down asynchronous counter- shift registerSerial in serial out-serial in parallel out-parallel in serial out - parallel in parallel out-applications.

## MODULE IV

Analog to digital converter -various analog to digital converters-operation of single slop A/D converter-Successive approximation A/D converter-list some A/D ICs- digital to analog converter the operation of R- 2R D/A converter -binary weighted converter- list some D/A ICs-memory in digital instrument-types of memory-define Byte, nibble, capacity- various RAM- various ROM-compare ROM & RAM-display in digital instrument - LED,LCD,NIXIE TUBE- the specifications of digital meter display-display digit and counts-define resolution of digital meter-Sensitivity.

## REFERENCES


1. Floyd and Jain , Digital Fundamentals , Pearson Education
2. Anandkumar.A , Fundamentals of Digital circuits ,PHI
3. Morismano , Digital Design , PHI
4. Malvino,Leach , Digital Principles and applications , TMH
5. R.S Sedha , Electronic measurements and Instrumentation , S Chand and Co